Cars, chargers and simultaneity

Network effects in green technology adoption

Reading. Li et al. (2017), with the note *The Microeconomic Foundation for Li et al. (2017)* as the toolbox. Section, table and equation numbers are the paper's own.

Part 1 · Free entry 15 min

Why does the network grow with the fleet?

The paper's appendix derives the charging station equation from a model of free entry. This part goes through that derivation. Each charging station is one firm.

What you need. In the short-run equilibrium from the lecture, profit per station is $\pi_t = Q_t f(N_t)$, where Q_t is the stock of electric vehicles (EVs) and N_t the number of stations. A firm that enters in period t pays the fixed cost F_t once and earns π_s in every period $s \geq t$. Future payments are discounted with the factor $R \in (0, 1)$.

	t	$t+1$	$t+2$	⋯
Enter now	$-F_t$, π_t	π_{t+1}	π_{t+2}	⋯
Wait	0	$-F_{t+1}$, π_{t+1}	π_{t+2}	⋯

1. *When to enter.*
   (i) Write the present value, seen from period t, of each of the two choices in the figure.
   (ii) With free entry, a firm is indifferent. Show that $F_t - RF_{t+1} = Q_t f(N_t)$.
   (iii) What does each side of this condition say about waiting?

2. *The station equation.* Add the subscript m for the market, and assume

$$\ln(F_{m,t} - RF_{m,t+1}) = \rho_0 + \rho_1' Z_{m,t} + \tau_{m,t},$$
$$\ln(f(N_{m,t})) = \lambda_0 + \lambda_1 \ln(N_{m,t}) + \vartheta_{m,t}.$$

   Solve for $\ln(N_{m,t})$. What are γ_0, γ_1 and γ_2 in the paper's equation (5)?

3. *The sign.* In the lecture, $f(N) = D(p^*)\varphi(N)/N$, where $\varphi(N)$ is the markup and $D(p^*)$ an EV owner's total charging demand.
   (i) Which sign must λ_1 have for $\gamma_1 > 0$?
   (ii) What does that sign say about competition between stations?

Stop point

The entry condition, γ_1 in terms of λ_1, and the sign of λ_1.

If you finish early

One of the equations (A1)–(A4) in the paper's appendix has an error. Which one, and how do you know? Then use the paper's estimate of γ_1 (Table 7, column e) to find λ_1. What does it say about the effect of one per cent more stations on a station's profit per EV?

Part 2 · Simultaneity 15 min

Which way does simultaneity bias OLS?

Sales and stations rise together, and each of the paper's estimating equations (4) and (5) has the other's dependent variable, or its stock, on the right. This part asks which way that biases least squares. The model is our own simplification and is not in the paper.

What you need. A stochastic version of the two equations, without controls and fixed effects, and with $\delta = 0$, so that the stock equals this period's sales, $Q = q$:

$$\ln(q) = \beta \ln(N) + \epsilon \qquad \text{(EV sales, with a demand shock } \epsilon\text{)}$$
$$\ln(N) = \gamma \ln(q) + u \qquad \text{(stations, with a shock } u \text{ to station investment)}$$

where $\beta, \gamma > 0$ play the roles of β_1 and γ_1, and $\beta\gamma < 1$. The two shocks have mean zero, variances σ_ϵ^2 and σ_u^2, and are uncorrelated.

1. *The loop.*
   (i) A positive demand shock, $\epsilon > 0$, hits. Trace it through the two equations: what happens to sales, then to stations, then to sales again?
   (ii) Why do the rounds die out?

2. *The reduced form.* Write $\ln(q)$ and $\ln(N)$ as functions of the two shocks.
   Hint: insert the station equation in the sales equation.

3. *The bias.*
   (i) Find $\mathrm{Cov}(\ln(N), \epsilon)$. What is its sign?
   Hint: for constants a and b, $\mathrm{Cov}(aX + bY, X) = a\,\mathrm{Var}(X) + b\,\mathrm{Cov}(X, Y)$.
   (ii) OLS of $\ln(q)$ on $\ln(N)$ gives $\operatorname{plim} \hat{\beta}_{OLS} = \beta + \mathrm{Cov}(\ln(N), \epsilon)/\,\mathrm{Var}(\ln(N))$. Find $\mathrm{Var}(\ln(N))$. Does OLS over- or underestimate β?
   Hint: $\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$.
   (iii) Explain the direction in words.

Stop point

$\ln(N)$ as a function of ϵ and u, the sign of $\mathrm{Cov}(\ln(N), \epsilon)$, and the direction of the bias in $\hat{\beta}_{OLS}$.

If you finish early

Do the same for the station equation: find the bias of OLS of $\ln(N)$ on $\ln(q)$. Which shock now plays the role of ϵ? When is the bias small, and what does that say about the kind of instrument the paper needs for Q?